GATE

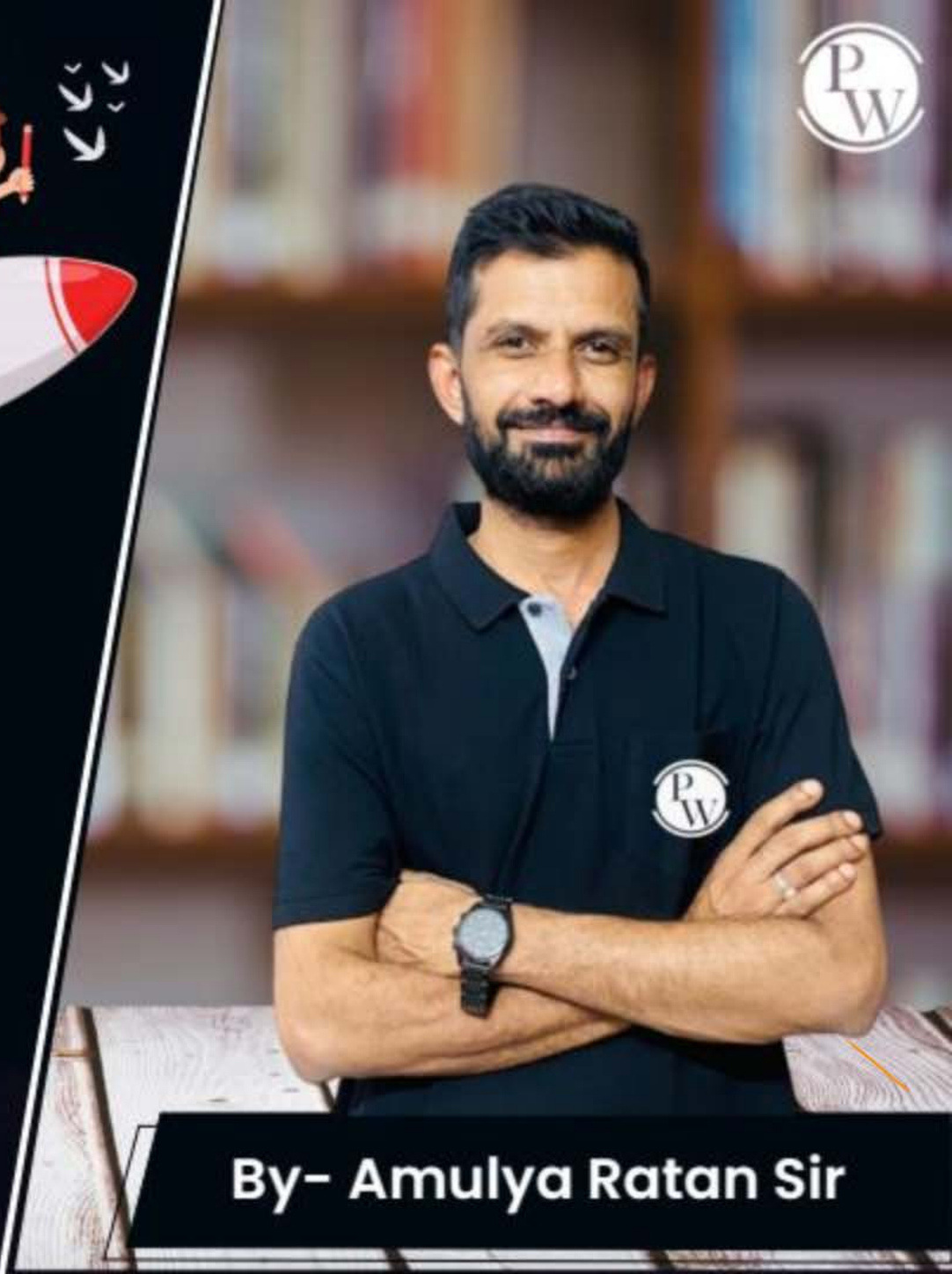
ALL BRANCHES



General Aptitude

QUANTITATIVE APTITUDE

Lecture No.- 12



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Recap of Previous Lecture



Topic Pipes and Cistern

Topics to be Covered



Topic-1 Time and Distance

[MCQ]

#Q. Pipes A and B can fill a tank in 4 hours and 8 hours respectively. Pipe C can empty it in 16 hours. If all the three pipes are opened together, then how long will it take to fill the tank?

Assignment

$A = \frac{1}{4} \quad B = \frac{1}{8} \quad C = \frac{-1}{16}$

$A\&B\&C = \frac{1}{4} + \frac{1}{8} - \frac{1}{16} = \frac{4+2-1}{16} = \frac{5}{16}$

$\frac{16}{5} = 3.2 \text{ hrs}$

A. 4.5 hours

B. 2 hours

C. 6 hours

D. 3.2 hours

[MCQ]

#Q. A tank can be filled by 20 buckets each of capacity 13.5 litres. If the capacity of each bucket be 9 litres, how many buckets will fill the same tank?

Assignment

A 30

B 25

C 20

D 15

$20 \times \frac{13.5}{9}$

$= \frac{2 \times 13.5}{9}$ (15)

$= 30$

Tank Cap $\Rightarrow 20 \times 13.5$

$= 270$ l

No. of Buckets $= \frac{270}{9} = 30$

No. of Bucket	Cap
20	13.5
?	9

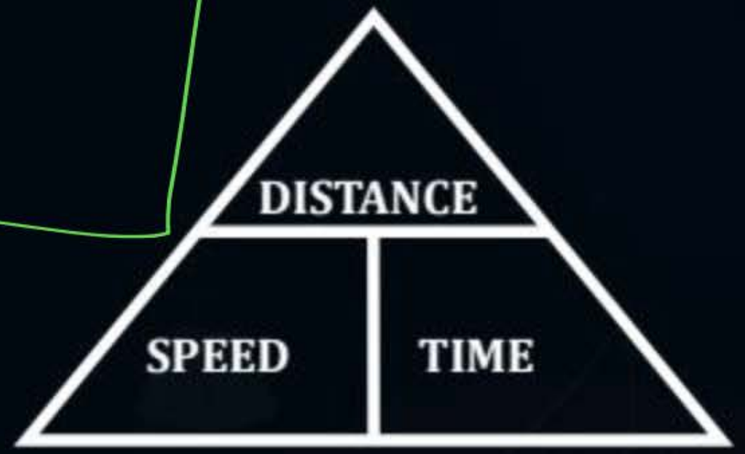
Time & Distance

$$D = S \times T$$

$$\text{Speed} = \frac{\text{Dist}}{\text{Time}}$$

$$T = \frac{D}{S}$$

$$\frac{\text{km}}{\text{hr}} = \frac{1000\text{m}}{3600\text{sec}} = \frac{5}{18}$$



* Relative Speed

* Average Speed

$$\text{km/hr} \xrightarrow{\times \frac{5}{18}} \text{m/sec}$$

$$\text{m/sec} \xrightarrow{\times \frac{18}{5}} \text{km/hr}$$

$D = 400\text{m}$

$S = 40\text{km/hr}$

$$T = \frac{400\text{m}}{40\text{km/hr}} = ?$$

[MCQ]

$T = \frac{D}{S}$

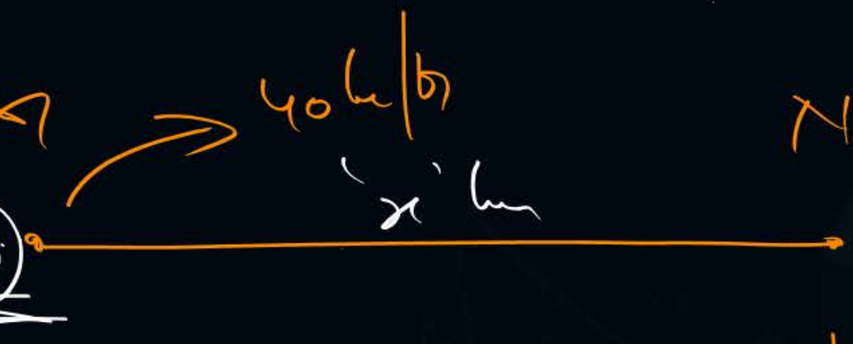
#Q. If you travel from M to N at 40km/hr and N to M at 60 km/hr. What would be your average speed of the journey?

$\frac{2xy}{x+y}$

$\frac{x}{40} + \frac{x}{60} = \frac{2x}{A.S.}$

$\frac{1}{24} = \frac{2}{A.S.}$

$\therefore A.S. = 48 \text{ km/h}$

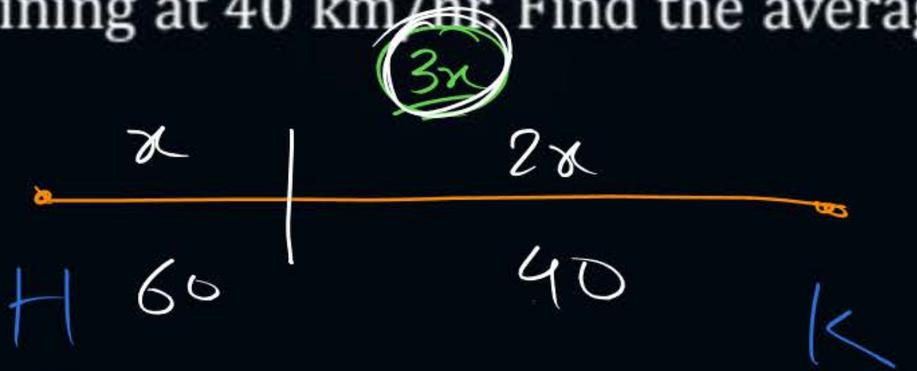


$A.S. = \frac{T.D.}{T.T.}$

$T.T. = \frac{T.D.}{A.S.}$

[MCQ]

#Q. A man travels from Howrah to Kanpur in his car. If he covers 1/3rd of the distance at 60 km/hr & remaining at 40 km/hr. Find the average speed of his journey.



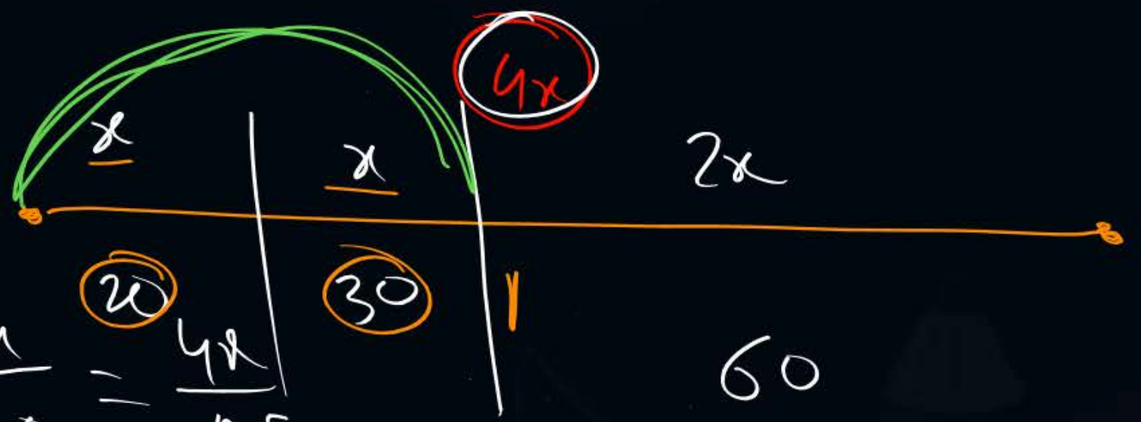
$$\frac{x}{60} + \frac{2x}{40} = \frac{3x}{A.S.}$$

$$\frac{2x+6x}{120} = \frac{3x}{A.S} \Rightarrow \frac{8x}{120} = \frac{3x}{A.S}$$

$$\therefore A.S. = 45 \text{ km/hr}$$

[MCQ]

#Q. A man travels ¼th of his journey at 20 km/hr, another ¼th at 30 km/hr & remaining at 60 km/hr. Find the average speed of his journey.



$$\frac{x}{20}+\frac{x}{30}+\frac{2x}{60}=\frac{4x}{A.S.}$$

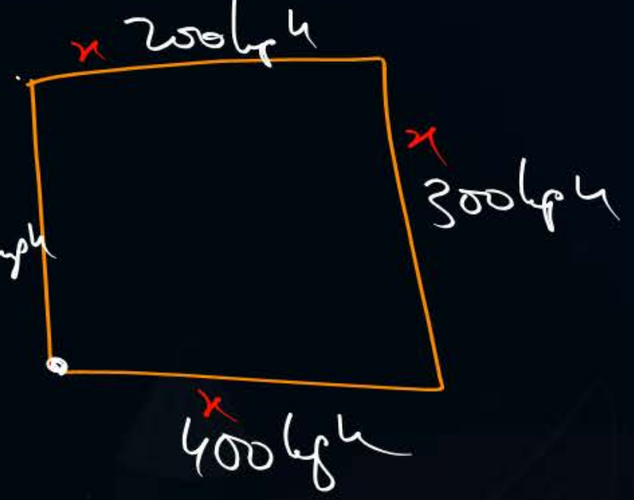
$$\frac{7x}{60}=\frac{4x}{A.S.}$$

$$\Rightarrow A.S.=\frac{60\times4}{7}=\frac{240}{7}=34.28$$

$$\frac{2\times24\times30}{54}=\frac{2\times24\times60}{84}=\frac{480}{7}$$

[MCQ]

#Q. If a car travels along four sides of a square at 100 kmph, 200 kmph, 300 kmph and 400 kmph. Find its average speed.



A. 152 km/hr

B. 180 km/hr

C. 192 km/hr

D. 210 km/hr

$$\frac{4x}{A.S} = \frac{x}{100} + \frac{x}{200} + \frac{x}{300} + \frac{x}{400}$$

$$\frac{12x + 6x + 4x + 3x}{1200} = \frac{4x}{A.S.}$$

$$\Rightarrow \frac{25x}{1200} = \frac{4x}{A.S}$$

$$48$$

$$\therefore A.S. = 48 \times 4$$

$$= 192$$

[MCQ]

$\frac{x}{2\times60} + \frac{x}{4\times30} + \frac{x}{4\times10} = \frac{x}{A.S.}$

#Q. A tourist covers half of his journey by train at 60 km/hr, half of the remainder by bus at 30 km/hr and the rest by cycle at 10 km/hr. Average speed of the tourist during the journey is

$\frac{2x}{2\times60} + \frac{4x}{4\times30} + \frac{4x}{4\times10} = \frac{4x}{A.S.}$



$\frac{2x}{60} + \frac{x}{30} + \frac{x}{10} = \frac{4x}{A.S.}$

$\frac{10x}{60} = \frac{4x}{A.S.}$

$\therefore A.S. = 24 \text{ km/hr}$

A. 36 km/hr

B. 33 km/hr

C. 24 km/hr

D. 18 km/hr

[MCQ]

#Q. A person covers a certain distance at a certain speed. If he decrease his speed by 20%, then he takes 16 minutes more to cover the same distance. Find the time taken by him to cover the original distance at original speed.

$\therefore O \cdot T = 16 \times 4 = 64 \text{ min}$

$\frac{100}{80} O \cdot T - 1 O \cdot T = 16$

$\Rightarrow \left(\frac{100}{80} - \frac{80}{80}\right) O \cdot T = 16$

$\frac{1}{4} \frac{20}{80} O \cdot T = 16$

$D = S \times T$

$S \propto \frac{1}{T}$ (constant D)

$\frac{x}{y} \propto \frac{y}{x}$

$\frac{80}{100} \quad \frac{x}{y} \quad \frac{100}{80}$

[MCQ]

#Q. Walking 3/4th of the usual speed a man is late to office by 10 minutes, find the usual time.

$\frac{4}{3} U \cdot T - U \cdot T = 10$

$S \propto \frac{1}{T}$

$\Rightarrow \frac{1}{3} U \cdot T = 10$

$U \cdot T = 30$ minutes

A 40 minutes

B 12 minutes

C 60 minutes

D 30 minutes

[MCQ]

$\frac{0.8}{90} = \frac{8}{900}$ 0.8 km

#Q. Mumbai police observes a thief 800 metres away from him. The thief started running at 80 km/hr & the police at 170 km/hr. In how much time the police can catch the thief?

$D = 800\ m$

$R.S. = 90\ km/hr \times \frac{5}{18} = 25\ m/sec$

$T = \frac{800\ m}{25\ m/sec} = 32\ seconds$

A 8.88 minutes

B 8.88 hours

C 32 seconds

D 32 minutes

[MCQ]

$T = \frac{D}{S}$

$\frac{x}{60} - \frac{x}{80} = \frac{15}{60}$ hr $= \frac{1}{4}$

#Q. An employee goes to his office from his house at a speed of 60 km/hr and reaches his office 10 minutes late. If he follows the speed of 80 km/hr, he reaches his office 5 minutes early. Find the distance he covers to reach his office.

A 55 km

B 45 km

C 50 km

D 60 km

$D = 60 \text{ km}$

$D = S \times T$

$\Rightarrow 60 \times \left(T + \frac{10}{60}\right) = 80 \times \left(T - \frac{5}{60}\right)$

$T = \frac{5}{6}$

$\frac{x}{240} = \frac{1}{4}$

$\frac{x}{60} = 1$

$S = 60 \times \frac{6}{5}$

$= 72 \text{ kmph}$

10:00 AM

60 kph $\Rightarrow$ 10:10

80 kph $\Rightarrow$ 9:55

[MCQ]

#Q. A man misses a train by 40 minutes if he travels at 30 kmph. If he travels at 40 kmph, then also he misses the train by 10minutes. What is the minimum speed required to catch the train on time?

Assignment

A 50 km/hr

B 55 km/hr

C 60 km/hr

D 45 km/hr

[MCQ]

#Q. A car driver driving at a speed of 68 km/hr locates a lorry 40 meters ahead of him. After 10 seconds, the lorry is 60 meters behind him. Find the speed of the lorry.

Assignment

[MCQ]

Assignment

#Q. Two boys begin together to write out a booklet containing 817 lines. The first boy starts with first line, writing at the rate of 200 lines an hour and the second boy starts with last line. He writes line 817 and so on backwards proceeding at the rate of 150 lines an hour. At what line will they meet?

A 469th

B 467th

C 465th

D 463rd

Meeting at a point:

$D_1 + D_2 = T.D.$

$D_1 = D_2$



Meeting at a point:

Two objects moving in opposite directions towards each other:
Distance covered by 1st object + Distance covered by 2nd object = Total distance





Two objects moving in same directions :
Distance covered by 1st object = Distance covered by 2nd object





[MCQ]

Assignment

#Q. A train starts from Ahmedabad to Howrah at 5am everyday with the speed of 60kmph. Another train starts from Howrah to Ahmedabad at 7am everyday with the speed of 80 kmph. If the distance between Ahmedabad and Howrah is given as 820km, then at what time the two trains meet each other?

[MCQ]

Assignment

#Q. A train starts from Chennai to Delhi at 11am with the speed of 110kmph. Another train starts from Chennai to Delhi at 9am the same day with the speed of 90 kmph. If the distance between Chennai and Delhi is given as 1920km, then at what time the two trains meet each other?



2 mins Summary

Topic Time & Distance

THANK - YOU